

packages/graph-core/src/analyzer/layer-detector.ts

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Overview

`packages/graph-core/src/analyzer/layer-detector.ts` adds a lightweight layer detection system and LLM-driven layer assignment for knowledge graphs. The file introduces new types, heuristics, and helper functions that operate on `KnowledgeGraph` nodes.

Key changes

- **Import** (line1): `import type { KnowledgeGraph, Layer } from "../types.js";`.
- **LLM response contract** (lines6-10): `export interface LLMLayerResponse` defines `name`, `description`, and `filePatterns`.
- **Heuristic detection**
 - `LAYER_PATTERNS` (lines16-67) maps directory patterns to layer names.
 - `matchFileToLayer` (lines80-97) assigns a file to the first matching pattern.
 - `detectLayers` (lines105-144) builds a `Layer[]` from graph nodes, defaulting unmatched files to a "Core" layer.
- **LLM prompt & parsing**
 - `buildLayerDetectionPrompt` (lines150-169) generates a prompt listing file paths.
 - `parseLayerDetectionResponse` (lines177-219) extracts and normalizes JSON from LLM output, handling code fences.
- **LLM-based layer application** (`applyLLMLayers`, lines227-284) assigns file nodes to layers based on `filePatterns`, creating an "Other" layer for unmatched files.

Impact

- **Correctness:** Adds deterministic layer assignment; heuristic may misclassify edge cases but defaults to "Core".
- **Maintainability:** Centralizes layer logic; future pattern tweaks can be made in `LAYER_PATTERNS`.
- **Performance:** $O(n \times m)$ where n is nodes and m is patterns; acceptable for typical graph sizes.
- **Compatibility:** No API changes to existing modules; purely additive.
- **Observability:** New functions can be unit-tested; LLM prompt generation is deterministic.

Risks & follow-ups

- **LLM output format:** `parseLayerDetectionResponse` expects a JSON array; verify against actual LLM responses.
- **Pattern coverage:** Missing or overlapping patterns may mislabel files; run integration tests on diverse codebases.

- **"Other" layer handling:** The layer is created lazily; nodes with no matching pattern are added to "Other".
- **Path normalization:** Windows backslashes are replaced, but relative paths may still misalign; validate with mixed-OS projects.